

1. Loop structures are used to repeat blocks of code without having to manually type it again.
2. The difference is a while statement will check the condition before running the code, so there's a chance it might not run at all. A do-while statement will run the code first and then check the condition after, ensuring it runs at least once.
4. a) An infinite loop is a loop that never stops running because the condition is always true.
b) The loop condition is always true, the loop is not updated properly.
c) Overflow is when a value exceeds the maximum limit that that variable can hold, causing incorrect results.
5. 60 times.
6. Any value of x less than 123 will make the loop infinite.
7. A counter keeps track of how many times something happens, an accumulator keeps a running total by adding values.

Uses of a counter: Counting how many times a loop runs, counting the number of times.
Uses of an accumulator: Finding the total sum of numbers, calculating total scores.

```
8. int total = 0;

   for (int i = 3; i <= 10; i++) {

       total += i;

   }
```

9. Number of iterations known or unknown: Use a for loop when you know the specific amount of times the program is supposed to be run. Use a while loop when the number of iterations is unknown.

Condition for stopping the loop: Choose a loop based on how the loop will end.

11. a) 10
b) "my"
c) "my string."
d) "MY STRING"
e) "my string."